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# Frequently Asked Questions

**Open Network for Data Exchange & 1864 Prep**

ATTACHMENT 4 · FAQ

The questions agencies, funders, and data-protection reviewers ask most about the Open Network for Data Exchange and 1864 Prep — with direct answers.

**Sections**

1. The basics
2. Data protection and trust
3. Joining and running it
4. Standards and interoperability
5. Cost, status, and next steps

## 1. The basics

### What is the Open Network for Data Exchange?

It is an open, decentralised coordination layer that lets one government agency ask another a verified question about a person or a case — and get a signed answer in seconds — without any agency surrendering control of its data. There is no central database. Each agency answers from its own records, signed with its own key.

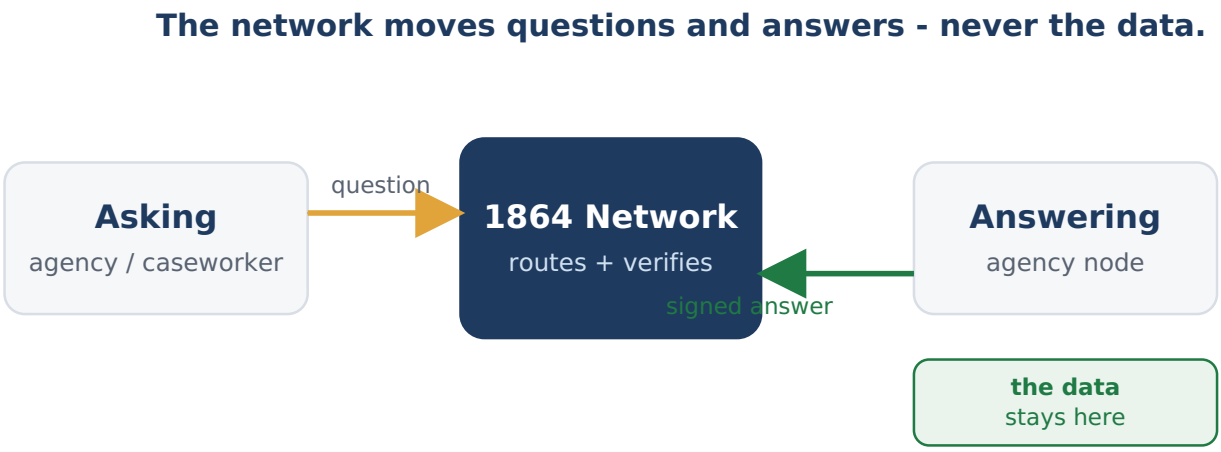



*Figure: only a signed answer travels; each agency keeps its records.*

### How is this different from a data warehouse or a master database?

A warehouse copies everyone’s data into one place, which concentrates risk and requires agencies to give up control. The Open Network does the opposite: data stays where it is, and only a specific authorised answer travels. Nothing is centralised except the directory of who is who and their public keys.

### What is 1864 Prep, and how does it relate to the network?

1864 Prep is a free, open-source tool that cleans and standardises an agency’s data on the agency’s own computer. It is the groundwork: an agency cleans its file once, locally, and is then ready to answer questions on the network. The two are designed to work together, but each is useful on its own.

### Is any of this actually built, or is it a concept?

Both pieces are built. 1864 Prep is live and open source and can be tried in a browser today. The Open Network runs end to end in prototype: three real Nigerian agency identities (NPC, NIMC, NHIA) are registered with their own keys and answer verified queries under their own identities, with each answer verifying successfully.

## 2. Data protection and trust

### Does 1864 hold or see our data?

No. 1864 operates the registry and the routing — the directory of participants and the switchboard that carries questions. It does not hold, store, or read the data agencies keep. Answers are produced by each agency's own node and signed with a key only that agency holds; 1864 cannot sign as an agency or pull its records.

### What actually leaves our building when we answer a question?

Only a signed answer to the specific authorised question — for example "registered", "NIN assigned", or "enrolled". Never the file, never the database, never a bulk export. The connector is built to return only the one scoped field required, not the whole record.

### How do you prevent someone impersonating an agency?

Every message is cryptographically signed with an Ed25519 key. When an answer arrives, the receiver looks up the sender's public key in the shared registry and verifies the signature. An answer whose signature does not match the registered key is rejected. Private keys are generated on each agency's own machine and never leave it.

### Is this compliant with Nigeria's data-protection law?

The architecture is designed to support compliance: data minimisation (only scoped answers move), purpose limitation and consent enforced at the policy layer, an audit trail that a query occurred, and local data custody. Because the operator is verifiably data-blind, an independent review can confirm the property rather than take it on trust; we welcome an NDPA review as part of a pilot.

### Who decides which questions our agency will answer?

The agency does. Participation is opt-in, and the specific questions an agency exposes are agreed at onboarding and enforced by its connector and the policy layer. An agency is never obliged to answer more than it has agreed to.

### What is logged, and who can see the logs?

The network records that a query happened — who asked, of whom, and when — for accountability, without recording the payload contents held by the agency. This supports audit and consent oversight while preserving data minimisation.

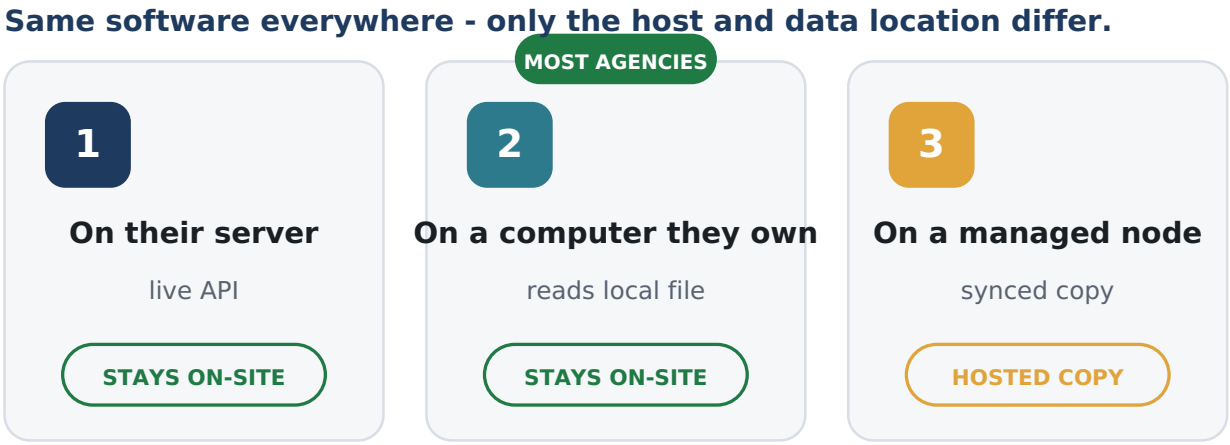
## 3. Joining and running it

**Our agency only has spreadsheets and no IT team. Can we still join?**

Yes — this is exactly who the network is built for. A small node is placed on a computer the agency already owns; it reads the spreadsheet locally and answers on the agency's behalf. Staff keep using Excel exactly as before. The only commitment is keeping that computer on and online during the hours the agency wants to be reachable.

**What are the ways to connect?**

Three, all running the same software: (1) on an agency's existing server, calling its live API; (2) on a computer the agency owns, reading a local spreadsheet or file — the most common case; (3) on a managed node hosted by 1864 reading a synced copy, used only for non-sensitive registers. In the first two, data stays on-site.




**Do we need to buy hardware?**

Usually not — the software runs on a computer the agency already has. Where there is no suitable always-on machine, a small pre-configured device can be provided. Where mains power is unreliable, a UPS or inverter covering the active hours is recommended.

**What happens if our computer is switched off or the internet drops?**

While the node is offline, the agency simply is not answering; nothing breaks, and it resumes when the machine is back on and connected. Agencies set the hours they want to be reachable, and that availability window is part of what the network advertises about them.

**How long does onboarding take?**

The largest piece of per-agency work is writing the connector that maps the agency's specific columns or API to the standard question format, plus registering its identity and admitting it to the network. For a spreadsheet agency this is typically a short engagement; agencies with a clean API are faster still.

## 4. Standards and interoperability

### What is this built on? Are we locked into a vendor?

It is built on Beckn, the open protocol behind India's national networks, and the field semantics follow NGDX, Nigeria's own Government Data Exchange standard. Both are open. There is no proprietary lock-in; joining means adopting standards that are already public and, in the case of NGDX, already endorsed by the country.

### How does this relate to NGDX?

NGDX is the national data rail that moves verified records between federal agencies. The Open Network is a layer above it that lets any organisation discover a service and transact, and can call NGDX to verify a federal fact with consent. It runs with or without NGDX present, so agencies — including states, LGAs, clinics, and NGOs — can begin coordinating now.

### Can non-government organisations join?

Yes. Admission is through the registry the operator runs, so clinics, fintechs, and NGOs can participate on the same terms as government agencies, subject to the same identity, consent, and authorisation rules. This is what makes it an open network rather than a closed government intranet.

### Does an agency have to change its internal systems?

No. The whole point of the connector is that the agency keeps its existing system — spreadsheet, database, or API — and the node translates between that system and the network. Joining the network is not a migration project.

## 5. Cost, status, and next steps

### What is the ask, and what are the next steps?

To fund a first pilot in a single sector with a lead agency, formalised by a memorandum of understanding, over about six months, with independent evaluation by CEGA at UC Berkeley. Immediate next steps we would welcome: a live demonstration of the birth loop, a technical deep-dive, and a conversation about scope and partners.

### How can we try it right now?

Open 1864 Prep at one864prep-demo.onrender.com and run a messy file through it — no installation, synthetic data only. The source is at github.com/abiola1864/1864Prep. For the network, we would set up a short live demonstration of the three-agency birth loop at your convenience.

### What would funding actually pay for?

Five things: agency onboarding at scale (installs, nodes, connectors, and field power such as UPS units); hardening 1864 Prep for more formats and registers; network operations (running the registry, routing, identity keys, and monitoring); security and governance (key custody, the consent and audit layer, and the NDPA review); and the pilot itself with its independent evaluation.

### Why is 1864 the right operator?

The value of the network is not the protocol — Beckn and NGDX are open. It is that someone must absorb the operational burden so that even a spreadsheet-only agency can join: building connectors, running nodes, issuing and safeguarding identity keys, and keeping the network running. That operator role — neutral, non-data-holding, and focused on reaching the least-digitised — is what 1864 is built to do.

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For more detail see the **Open Network User Guide**, the **1864 Prep User Guide**, and the **Technical Documentation**. Contact: abiola@1864institute.org · 1864institute.org.